Introduction to Economics

Review for Midterm 1

Blocks 1–4

Pavel Kocourek · Universitat de Barcelona · Monday 16 November 2026

Plan for today

- Midterm 1: the paper and the rules
- Part A: six kinds of question, in pairs
- Part B: the six graph questions
- Three Part B pages, solved on the board
- Thursday: what to bring

Slide 1 Title

Preview version. The full slides and handout are posted after class.

The review session for Midterm 1, Blocks 1 to 4. The first hour sets out exactly what Thursday's paper looks like, then goes through the six kinds of multiple-choice question and two graph moves, each tried first with a neighbour and then explained. After the break come three full graph pages in the practice exam's form, each tried alone and then solved on the board the way you would write it on Thursday. Every example is new.

Slide 2 Plan for today

The session in five topics. Before the break: the paper and its rules, then the six kinds of multiple-choice question of Part A and the six graph questions of Part B, two of them practised as single steps with a neighbour. After the break: three full Part B pages (a market drawn from a table with a shift and an elasticity, a tax with its areas, and a price-taking firm's costs and profit), each tried alone first and then solved on the board. The last slide repeats what to bring on Thursday.

Midterm 1 on Thursday: 6 multiple-choice and 6 graph questions

The paper

- Thursday 19 November 2026, 10:05–11:50 (105 minutes)
- Part A: 6 multiple choice, 24 points
- Part B: 6 graph questions on pre-drawn grids, 76 points
- About one point per minute

The rules

- Closed book: pen, paper and any handwritten notes (also photocopied or printed)
- No calculator: every number works out by hand
- No typed or printed text; no formula sheet is provided
- Phones off and away; photo ID on the desk
- Working supports answers; on a graph, the drawing counts as working

Slide 3 Midterm 1 on Thursday: 6 multiple-choice and 6 graph questions

Thursday's paper has the same length, format and rules as the practice exam on the course page, so one full sitting of it, on paper, is the best rehearsal. In Part A a wrong letter costs nothing, so never leave a question blank. In Part B each question has its own page: draw on the printed grid, label every line you draw, write the letters of an area inside it, each answer in its box and the working in the working space; a page's points tell you roughly how many minutes it deserves.

Read the notes rule carefully: notes must be in your own handwriting (a photocopy or printout of your handwritten notes is fine), typed or printed text is not, and no formula sheet is provided: copy any formula you want into your handwritten notes.

Opportunity cost: count what you give up — a sunk cost is never part of it

- Opportunity cost: the best thing you give up + money you spend only because you choose it
- Sunk cost: already paid whatever you choose — leave it out
- Comparative advantage: the lower opportunity cost
- Both gain from trade at any price between the two opportunity costs

Class 2, slide 23 (sunk cost) · class 3, slide 18 (comparative advantage).

Slide 4 Opportunity cost: count what you give up — a sunk cost is never part of it

Two of Part A's six kinds of question, both from classes 2 and 3. An opportunity cost asks what changes because you choose one option: the best alternative you give up, plus any money you spend only because you choose it. Money already paid that you cannot get back is the same whichever you choose, so it cannot help you decide: leave it out. For trade, each side's cost of one unit is counted in the other good it gives up; the side with the lower cost specialises in that good, and any price between the two costs leaves both better off, even when one side is faster at everything (absolute advantage).

Friday's circus show: what does going cost you?

Part A [4 points]

You paid 28 euros, no refund, for a circus show on Friday evening. Instead you could hand out leaflets: 4 hours at 13 euros an hour. The bus to the show and back costs 6 euros. Nothing else changes. What is your opportunity cost of going to the show?

- (A) 86 euros
- (B) 58 euros
- (C) 52 euros
- (D) 34 euros

Box 1

Answer

A B C D

In pairs, 2 minutes.

Slide 5 Friday's circus show: what does going cost you?

A Part A question on opportunity cost: a ticket already paid, an evening job you could do instead and a bus fare. Try it before Monday: for each of the three amounts, ask whether going to the show changes it, then add up only the amounts that it does.

Revealed in class.

Ironing or windows: who should iron, and at what swap?

Part A [4 points]

Two siblings share the housework. Hours each needs:

	one basket of ironing	one window washed
the older	1	¼
the younger	3	½

Both gain from which plan?

- (A) No swap: the older is faster at both.
- (B) The younger irons; swap 1 basket for 5 windows.
- (C) The older irons; swap 1 basket for 7 windows.
- (D) The older irons; swap 1 basket for 5 windows.

Box 2

Answer

A B C D

In pairs, 2½ minutes.

Slide 6 Ironing or windows: who should iron, and at what swap?

A Part A question on comparative advantage: two siblings, the hours each needs for a basket of ironing and for a window, and four plans. Try it before Monday: work out what one basket costs each sibling in windows not washed, decide who should iron, and ask which swaps leave both better off.

Revealed in class.

Demand moves for buyers' reasons; elasticity decides what a price rise does to revenue

- The good's own price changes: a move along its demand curve, never a shift
- Demand shifts with incomes, related goods' prices, tastes, expectations, the number of buyers
- Read backwards: price and quantity move the same way → demand moved; opposite ways → supply moved
- Price elasticity of demand = |% change in quantity ÷ % change in price|, each % of the midpoint
- Below 1 (inelastic): a higher price raises revenue; above 1 (elastic): it lowers revenue

Inferior good: bought less when incomes rise (class 5, slide 14) · midpoint method: class 7, slides 7–8.

Slide 7 Demand moves for buyers' reasons; elasticity decides what a price rise does to revenue

Two more Part A moves, from classes 5 and 7. A demand curve is drawn for given incomes, prices of related goods, tastes, expectations and the number of buyers: change one of those and the whole curve shifts, while a change in the good's own price only moves buyers along it. An inferior good is one that people buy less of when their incomes rise.

The same logic read backwards tells you which curve moved. A shift of demand takes the market along the supply curve, which slopes up, so the price and the quantity change in the same direction; a shift of supply takes it along the demand curve, which slopes down, so they change in opposite directions.

The price elasticity of demand compares the two percentage changes, each taken of the midpoint so that the answer is the same in either direction, and is reported as a positive number. Below 1 the quantity reacts less than the price, so a higher price brings in more revenue; above 1, less.

Incomes rise: what happens in the market for canned soup?

Part A [4 points]

Incomes in a city rise. Canned soup is an inferior good there. What happens in the market for canned soup?

(A) Demand shifts right.
(B) Demand shifts left; the lower price then shifts it back right.
(C) Demand shifts left; the price and the quantity fall.
(D) Supply shifts left: shops sell less soup.

Box 3

Answer

A B C D

In pairs, 1½ minutes.

Slide 8 Incomes rise: what happens in the market for canned soup?

A Part A question on what shifts demand: incomes rise, and canned soup is an inferior good. Try it before Monday: decide which curve moves and in which direction, and then what happens to the price and to the quantity sold.

Revealed in class.

A more expensive bowl of ramen: elastic or inelastic, and revenue?

Part A [4 points]

A ramen bar raises the price of a bowl from 9 to 13 euros. Bowls sold a day fall from 120 to 100. Which statement is correct for this price range?

- (A) Demand is inelastic, and total revenue rises.
- (B) Demand is inelastic, and total revenue falls.
- (C) Demand is elastic, and total revenue rises.
- (D) Demand is elastic, and total revenue falls.

Box 4

Answer

A B C D

In pairs, 2½ minutes.

Slide 9 A more expensive bowl of ramen: elastic or inelastic, and revenue?

A Part A question on elasticity and revenue: a ramen bar raises the price of a bowl and sells fewer bowls a day. Try it before Monday: compute the price elasticity of demand by the midpoint method, then revenue before and after, and check that the two tell the same story.

Revealed in class.

A monopoly sells while marginal revenue beats marginal cost; a Nash equilibrium is a pair of best replies

- Marginal revenue (MR): what one more unit adds to revenue — the price falls for all
- A monopoly sells every unit whose MR is above its marginal cost (MC)
- Its price: read on the demand row at that quantity
- After a cost rise: the same rule with the new MC
- Nash equilibrium: each player's choice is its best reply to the other's

Monopoly: the only seller, no close substitute (class 12, slide 20) · MR (class 13, slide 4) · Nash equilibrium (class 13, slide 12).

Slide 10 A monopoly sells while marginal revenue beats marginal cost; a Nash equilibrium is a pair of best replies

Two Part A moves from class 13. Marginal revenue (MR) is what one more unit adds to revenue; for a monopoly it is below the price, because selling one more means a lower price for every buyer, not only for the last. Work row by row: revenue = price × quantity, then what each step adds, per unit, and keep every step whose MR is above the marginal cost (MC); the price is then read on the demand row, never taken from the MR. A cost rise changes nothing in the method: the buyers are the same, so the revenue rows are the same, and only the MC you compare them with is new.

For a game, take each choice of the other player in turn and find your best reply to it; a Nash equilibrium is a cell where both players' best replies meet, so neither gains by changing alone.

The region's only zip line: which price maximises its profit?

Part A [4 points]

An adventure park runs the region's only zip line. Each extra ride costs it 17 euros (it has no fixed cost). The table gives the highest price at which it can sell each number of rides a day (one price for all riders). Which price maximises its profit?

Rides a day	100	200	300	400	500	600
Price (euros a ride)	35	33	31	29	27	25

(A) 25 euros (B) 19 euros
(C) 17 euros (D) 27 euros

Then: each extra ride costs 8 euros more. The new price?

Box 5

Answer

(A) (B) (C) (D)

In pairs, 2½ minutes.

Slide 11 The region's only zip line: which price maximises its profit?

A Part A question on a monopoly's price from a table: the region's only zip line, the highest price at which it can sell each number of rides a day, and a cost of 17 euros for each extra ride. Try it before Monday: compute revenue at each row, then what each extra hundred rides adds per ride, decide where to stop, and read the price from the right row. Then the follow-up under the options: each extra ride costs 8 euros more. Redo only the step that changes, find the new price, and compare its rise with the rise in the cost.

Revealed in class.

Two kebab shops, 8 euros or 6: where do they end up?

Part A [4 points]

A small town's only two kebab shops, Brasa and Flama, each charge 8 or 6 euros a kebab, at the same time and without talking. Profits, thousand euros a month:

Brasa's first		**Flama** 8 euros	**Flama** 6 euros
Brasa	8 euros	Brasa 12, Flama 9	Brasa 4, Flama 12
	6 euros	Brasa 16, Flama 2	Brasa 9, Flama 5

Which outcome is the Nash equilibrium?

(A) Brasa 8, Flama 8 (B) Brasa 8, Flama 6
(C) Brasa 6, Flama 6 (D) Brasa 6, Flama 8

Then: Brasa promises 8 and keeps it. What does Flama charge? What does Brasa earn?

Box 6

Answer

(A) (B) (C) (D)

In pairs, 2 minutes.

Slide 12 Two kebab shops, 8 euros or 6: where do they end up?

A Part A question on a game: two kebab shops, each charging 8 or 6 euros at the same time, and their monthly profits in a table. Try it before Monday: for each price of the other shop, find each shop's best reply, comparing only that shop's own numbers; then look for the cell where both best replies meet. Then the follow-up under the options: Brasa promises to charge 8 and keeps its word. Ask what Flama does then, looking only at Flama's own numbers, and what that leaves Brasa.

Revealed in class.

Part B: six graph questions, one move each

B1 Supply and demand from a table	plot, shift, E_1 and E_2, midpoint elasticity
B2 A tax	the tax line; quantity, two prices; CS, PS, T, DWL
B3 A price control	supplied, demanded, shortage or surplus; DWL
B4 Trade and a tariff	world price; production, consumption, imports; T, DWL
B5 An externality	SC or social value; efficient quantity; DWL; corrective tax
B6 A price-taking firm	MC and ATC from TC; best quantity; profit; lowest AVC

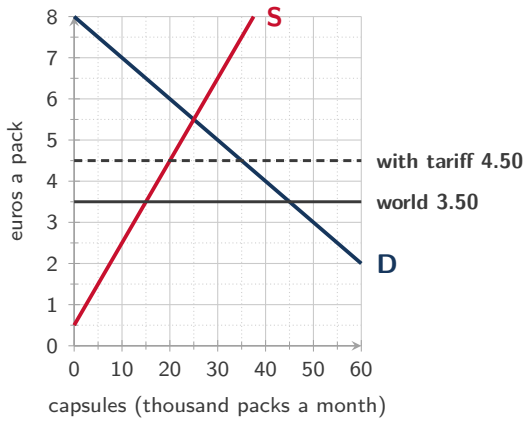
Now: B4 and B5 in pairs. After the break: B1, B2, B6 on the board.

D: demand · S: supply · E_1, E_2: the equilibria · CS: consumer surplus · PS: producer surplus · T: tax (or tariff) revenue · DWL: deadweight loss · SC: social cost · TC: total cost · ATC: average total cost (TC ÷ quantity) · AVC: average variable cost.

Slide 13 Part B: six graph questions, one move each

Part B of the practice exam has six graph pages, each built on one move you have drawn in class: a market from a table with a shift, a tax, a price control, trade with a tariff, an externality and a price-taking firm. The foot of the slide spells out the abbreviations used on them: consumer surplus (CS), producer surplus (PS), tax or tariff revenue (T), deadweight loss (DWL), social cost (SC), total cost (TC), average total cost (ATC) and average variable cost (AVC). On these grids every number you need sits on a solid or a dotted line, and the same answering rules hold on every page: label every line you draw, and write the letters of an area inside it, or beside it with an arrow when it is small. Next come the key moves of the tariff page and the external-cost page; after the break, three full pages.

A tariff on coffee capsules: what is collected, what is lost?



Coffee capsules (packs, thousand a month): a country's demand D and its producers' supply S. It imports at the world price of 3.50 euros a pack; then it puts a tariff of 1 euro on every imported pack.

With the tariff: the imports (thousand packs), T and DWL (thousand euros)?

Box 7

Imports	T	DWL

In pairs, 3 minutes.

T: tariff revenue = tariff × imports with the tariff · DWL: deadweight loss.

Slide 14 A tariff on coffee capsules: what is collected, what is lost?

The key move of the trade-and-tariff page, as three numbers: the grid prints the demand and the supply for coffee capsules, the world price and the price with the tariff. Try it before Monday: at the price with the tariff, read how many packs the country's producers make and how many buyers want, and find the imports; ask on which packs the tariff is paid, and find the area on the grid that equals the revenue; then find what is lost, the deadweight loss, as an area on the grid and as a number.

Revealed in class.

Patio heaters harm others: how many would be efficient?



Patio heaters (thousand a year): demand D and supply S. The gas each heater burns does 60 euros of harm to other people.

Box 8

Market quantity	Efficient quantity
DWL (thousand euros)	Corrective tax (euros a heater)

In pairs, 2½ minutes.

SC: social cost = sellers' cost + the harm to others, at every quantity · DWL: deadweight loss.

Slide 15 Patio heaters harm others: how many would be efficient?

The key moves of the external-cost page, as four numbers: the grid prints the demand and the supply for patio heaters, and the gas each heater burns does 60 euros of harm to other people. Try it before Monday on a sketch: find where the market ends up; draw the social-cost curve and decide how it sits against the supply curve; find the efficient quantity and the deadweight loss between the two quantities; then ask which tax would make sellers count the harm.

Revealed in class.

Break

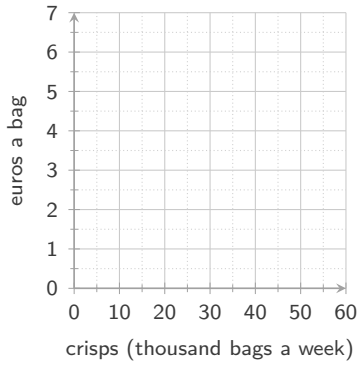
Ten-minute break

Slide 16 Break

Ten minutes. After the break: three full Part B pages, each tried alone and then solved on the board.

Crisps: two curves from a table, then a shift — five minutes alone

The market for crisps (thousand bags a week). The table gives the quantities demanded and supplied at three prices; both curves are straight lines. [12 points; drawing 6; boxes as marked]



Price (euros a bag)	2	3	4
Quantity demanded (thousand a week)	40	30	20
Quantity supplied (thousand a week)	10	20	30

(a) Draw the demand curve and the supply curve through the table's points, label them D and S, and mark the equilibrium E_1.

(b) The price of frying oil, an input used to make crisps, rises. At every price, sellers now offer 20 thousand fewer bags a week than before. Draw the new supply curve, label it S_2, and mark the new equilibrium E_2.

	Price (euros) [1.5]	Quantity (thousand) [1.5]
Box 9 (a) Before: E_1		
Box 10 (b) After: E_2		

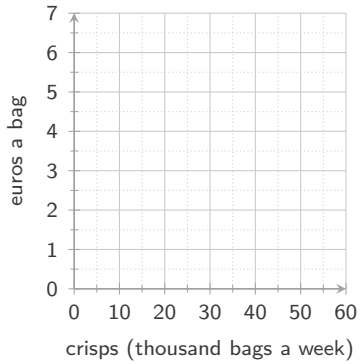
Slide 17 Crisps: two curves from a table, then a shift — five minutes alone

The first of three full Part B pages, in the practice exam's form: the market for crisps, a table of the quantities demanded and supplied at three prices, then a more expensive input. Try it before Monday on a sketch of the grid: read the scale on each axis, plot the table's points with the quantity across and the price up, draw and label both lines and their crossing; then draw the new supply curve and the new crossing, and ask which curve moved and which did not.

Revealed in class.

The crisps: the price elasticity of demand from E_1 to E_2 — two minutes alone

[2 points]



(c) Demand did not move, so E_1 and E_2 are two points on the demand curve. Compute the price elasticity of demand between them by the midpoint method, and write it as a positive number (a fraction or a decimal).

	Elasticity [2]
Box 11	

Price elasticity of demand: |% change in quantity ÷ % change in price|, each % of the midpoint.

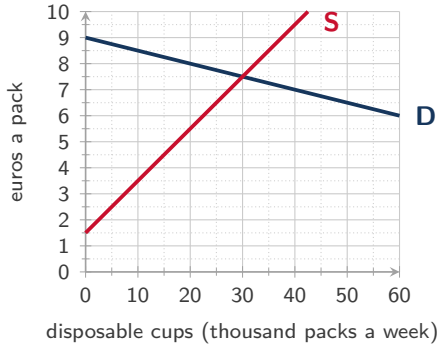
Slide 18 The crisps: the price elasticity of demand from E1 to E2 — two minutes alone

The same market: demand did not move, so the two crossings of the previous slide are two points on one demand curve, and the question asks for the price elasticity of demand between them by the midpoint method, as a positive number. Try it before Monday with your own two points: the change in quantity over the average quantity, the change in price over the average price, then one divided by the other, with each line of working written out.

Revealed in class.

A tax on disposable cups: the line and the prices — three minutes alone

The grid shows the market for disposable cups (packs, thousand a week): demand D and supply S. The state puts a tax of 2.50 euros on every pack sold; the sellers pay it to the state.



[5 points; drawing 2; boxes as marked]

(a) Draw the supply curve with the tax: at every quantity 2.50 euros above S. Label it S + tax.

(b) With the tax:

Box 12

Quantity sold (thousand) [1]

Buyers pay (euros a pack) [1]

Sellers keep (euros a pack) [1]

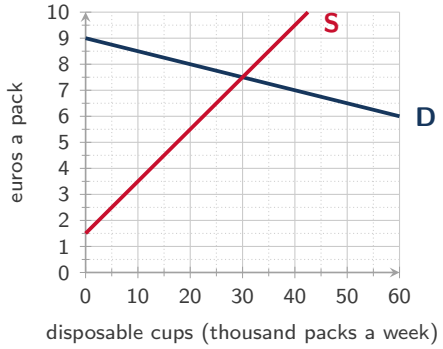
Draw on the grid and label every line you draw; numbers in the boxes.

Slide 19 A tax on disposable cups: the line and the prices — three minutes alone

The second full Part B page: the grid prints the demand and the supply for disposable cups, and the state puts a tax of 2.50 euros on every pack, paid by the sellers. Try it before Monday on a sketch of the grid: read the scale, draw the supply curve with the tax and label it, then find the quantity sold, the price buyers pay and the price sellers keep, with the working beside each.

Revealed in class.

The tax's areas as numbers and letters — and who bears more — three minutes alone



[9 points; letters 2; boxes as marked]

(c) With the tax, in thousand euros a week: CS, PS, T and DWL. Write each one's letters inside its area.

Box 13

CS [1.5] PS [1.5] T [1.5] DWL [1.5]

(d) Who bears more of the tax? Circle one. [1]

Box 14 (Buyers) (Sellers)

CS: consumer surplus · PS: producer surplus · T: tax revenue · DWL: deadweight loss. A triangle is ½ × base × height.

Slide 20 The tax's areas as numbers and letters — and who bears more — three minutes alone

The same market with the tax: consumer surplus, producer surplus, tax revenue and deadweight loss as numbers and as letters on the grid, and who bears more of the tax. Try it before Monday on your sketch from the previous slide: shade each area and write its letters inside it, compute each one (½ × base × height for a triangle), check that they add up to the surplus without the tax, then compare how far each side's price moved.

Revealed in class.

Carnations by the tray: the cost table — three minutes alone

A farm grows carnations and sells them by the tray at the market price of 20 euros a tray; it is a price taker. The table gives its total cost (TC) for 0 to 5 trays a day.

(a) Complete the table: the marginal cost (MC) of each tray (the extra cost of that tray) and the average total cost (ATC) at 2, 3 and 4 trays. [5.5 points; table cells as marked]

Trays a day	0	1	2	3	4	5
Total cost, TC (euros)	20	30	36	45	60	85
MC of that tray (euros) [0.5]	–	Box 15				
ATC (euros a tray) [1]	–	–				–

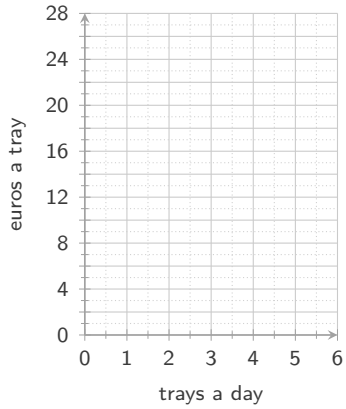
Box 16 What if the farm's rent rises by 10 euros a day?

MC of the 3rd tray: ____ ATC at 4 trays: ____

The farm on the grid: MC, ATC and profit — four minutes alone

The same farm, at the market price of 20 euros a tray. Its costs (euros):

[10.5 points; drawing 4.5; boxes as marked]



trays a day	0	1	2	3	4	5
TC	20	30	36	45	60	85
MC						
ATC						

(b) Draw MC as steps: over each tray, a horizontal line at that tray's MC. Mark the ATC at 2, 3 and 4 trays with dots and join them. Draw the price as a line labelled P.

(c) Best quantity (trays a day) [2] Profit (euros a day) [2]

Box 17 ____ ____

Shade the profit as a rectangle; write profit inside it.

(d) At 12 euros a tray: produce or shut down in the short run? Circle one; the reason in one line. [2]

Box 18 (Produce) (Shut down)

MC: marginal cost · ATC: average total cost · AVC: average variable cost, (TC − fixed cost) ÷ trays.

Slide 21 Carnations by the tray: the cost table — three minutes alone

The third full Part B page starts with a cost table: a carnation farm, a price taker selling at 20 euros a tray, and its total cost for 0 to 5 trays a day. Try it before Monday: fill the marginal-cost row, the extra cost of each tray, and the average total cost at 2, 3 and 4 trays; then ask what the cost at zero trays is, and which average leaves it out. Then the what-if under the table: the rent rises by 10 euros a day. Decide what kind of cost the rent is, and work out again the marginal cost of the third tray and the average total cost at 4 trays.

Revealed in class.

Slide 22 The farm on the grid: MC, ATC and profit — four minutes alone

The same farm on a grid, at the market price of 20 euros a tray. Fill the previous slide's cells first, then try it before Monday on a sketch of the grid: read the scale on each axis, draw the marginal cost as steps, the average total cost as joined dots and the price as a line, each with its label; find the best quantity and the profit, shade the profit, and decide what the farm does if the price falls to 12, with the reason in one line.

Revealed in class.

Before Thursday 19 November

- **Midterm 1**: Thursday 19 November, 10:05–11:50, in class (Blocks 1–4)
- Bring: a pen and your photo ID; handwritten notes if you want. No calculator.
- **Class 15**, Monday 23 November: gross domestic product (GDP) and the three ways to measure it
- Read before it: Mankiw & Taylor, Ch. 20, pp. 436–444 (sections 20.2, 20.3)
- Office hours by appointment: e-mail me



Good luck on Thursday!

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Slide 23 Before Thursday 19 November

Thursday 19 November: Midterm 1 on Blocks 1 to 4, in class; be in your seat a few minutes before the start, with a pen, your photo ID and, if you want them, your handwritten notes. The best preparation is the practice exam on the course page, done on paper in one sitting, giving each page about as many minutes as it has points. Monday 23 November: class 15 opens with gross domestic product (GDP) and the three ways to measure it; read Mankiw & Taylor, Chapter 20, pages 436 to 444 (sections 20.2 and 20.3). Office hours by appointment: e-mail me.

pkocourek.com/teaching/intro-economics · pkocourek@ub.edu